SSH Investigation Report

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Incident Analysis Findings

- **Initial Access Attempt:** Network packet captures indicate an initial unauthorized access attempt targeting the SSH service (port 22) on the Linux jump server on March 04.
- **Failed Authentication:** The logs confirm a failed password authentication event at 21:11:16 for an invalid or unauthorized user account named 'attacker'.
- **Source & Destination Identification:** The malicious login attempt originated from the external source IP 203.0.113.10, explicitly targeting the jump server's designated IP 10.0.0.50.
- **Successful Authentication Event:** Following the failed attempt, a subsequent successful login was recorded for the valid system user 'kali' originating from the same source IP (203.0.113.10) at 21:18:10.
- **Time Gaps & Patterns:** The timestamp delta between the failed access and the successful login attempt is approximately seven minutes. This significant gap indicates manual human interaction rather than an automated brute-force script, which would typically register milliseconds between attempts.
- **Protocol Analysis:** System logs were successfully extracted and encapsulated into Syslog UDP format for packet analysis, preserving the true historical chronological sequence of the authentication events.

Investigation Evidence

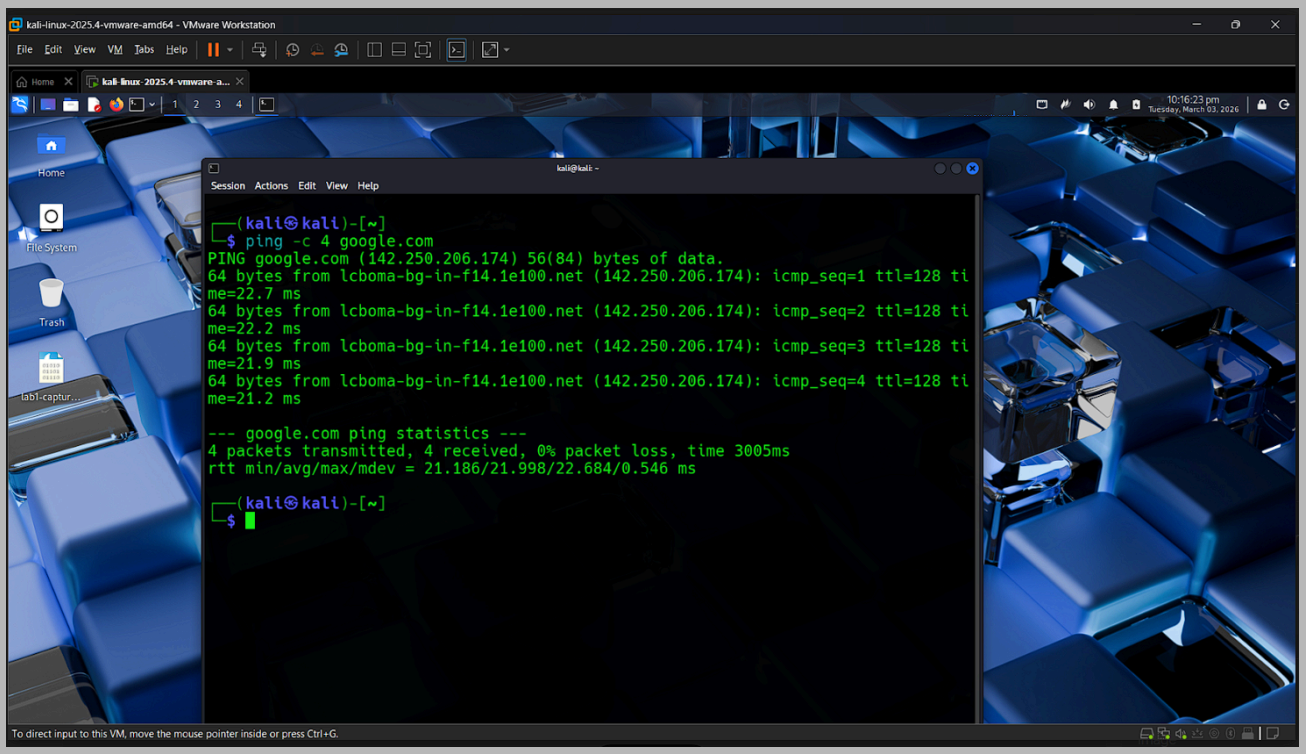

Figure 1: Kali Linux Network Verification

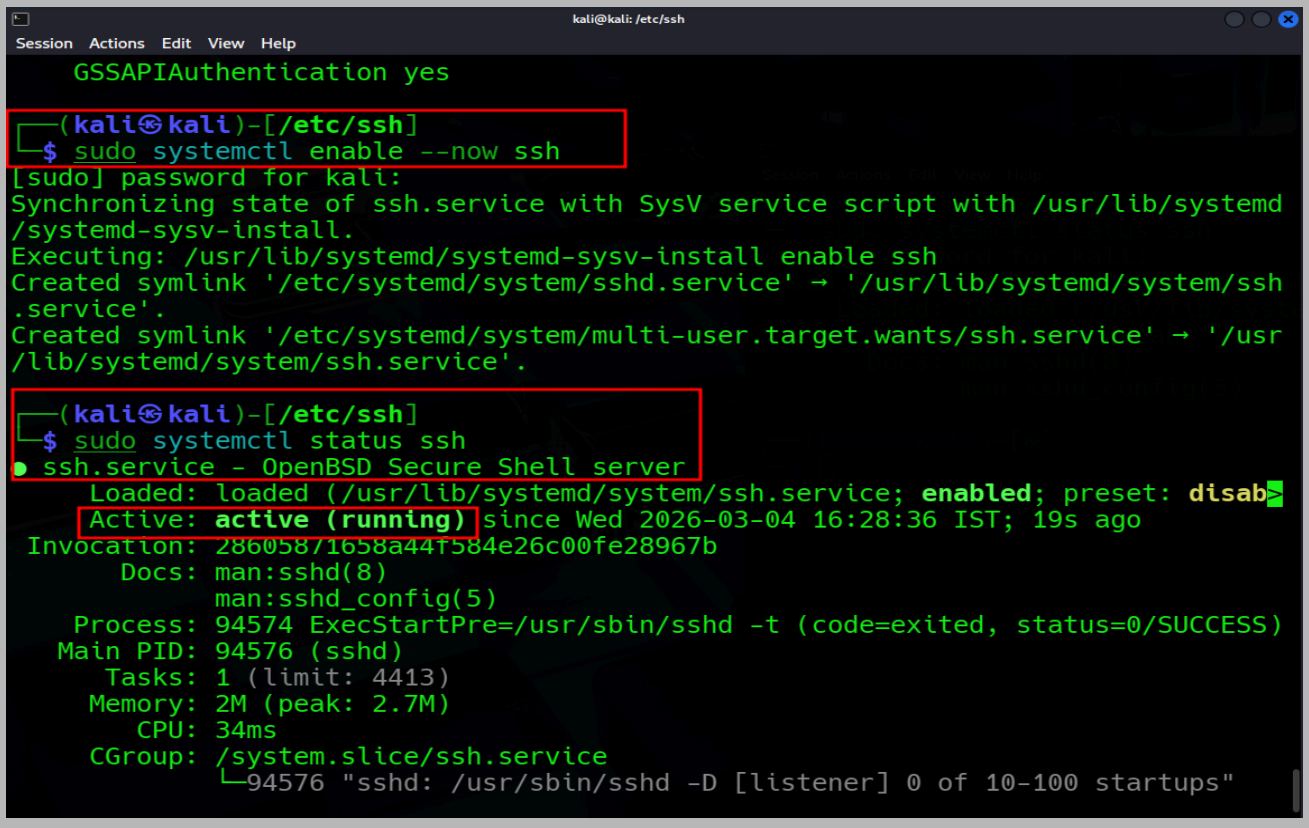

Figure 2: SSH Service Activation

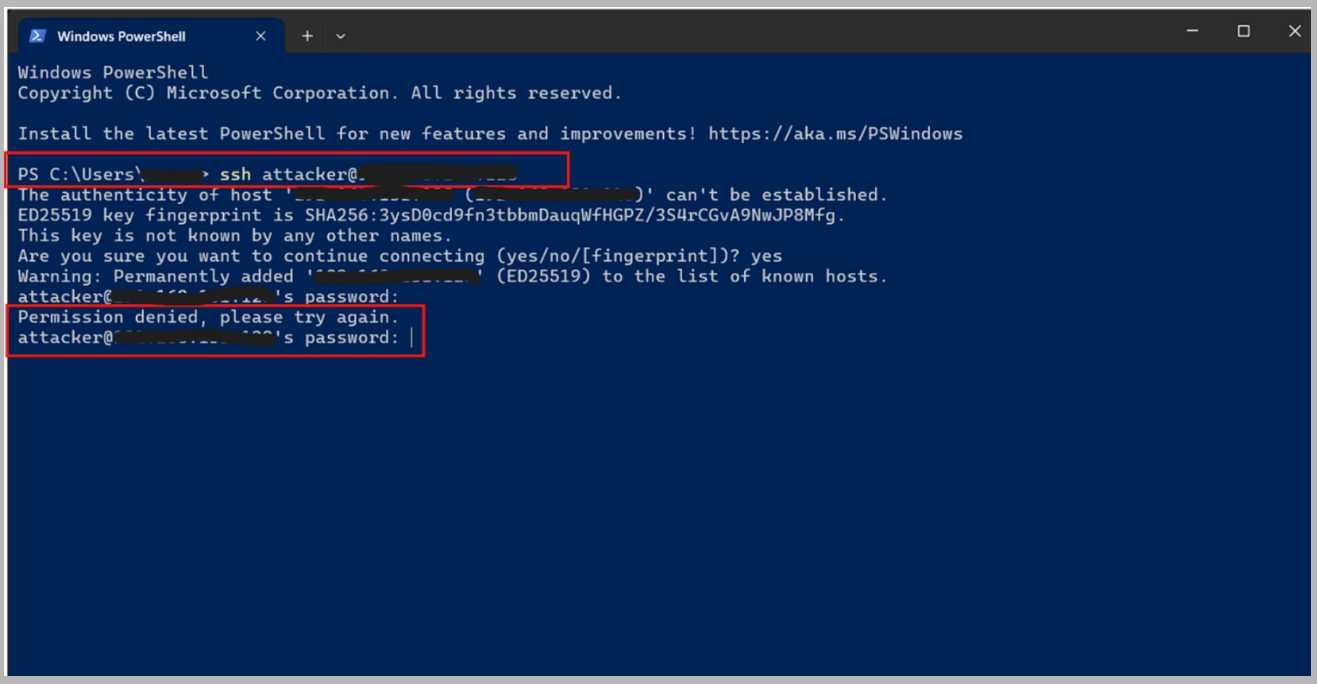

Figure 3: Failed Login Simulation

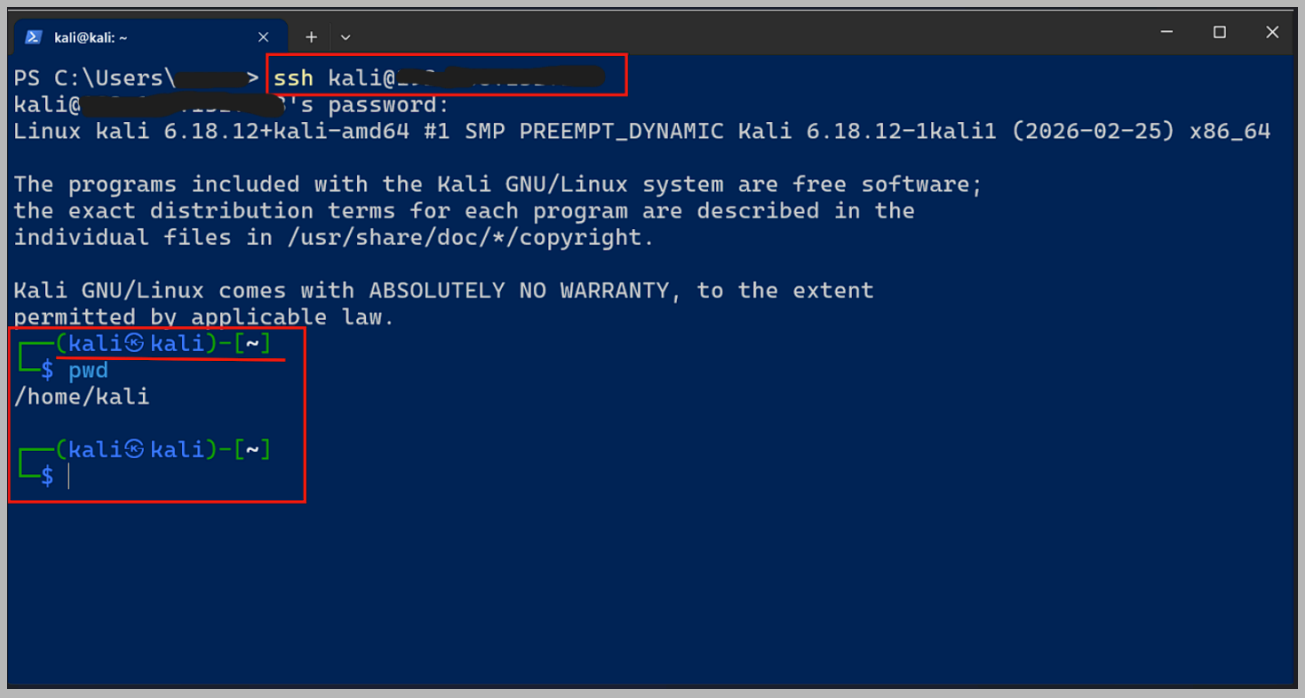

Figure 4: Successful Login Simulation

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/Desktop]
└─$ ls -la
total 6516
drwxr-xr-x  2 kali kali    4096 Mar  5 15:47 .
drwx------ 17 kali kali    4096 Mar  5 13:12 ..
-rw-rw-r--  1 kali kali    2170 Mar  5 15:47 advanced_pcap.py
-rw-rw-r--  1 kali kali     863 Mar  5 13:22 convert2pcap.py
-rw-r--r--  1 kali kali 6595848 Feb 21 13:14 lab1-capture.pcapng
-rw-rw-r--  1 kali kali   21932 Mar  4 22:19 my_ssh_logs.txt
-rw-r--r--  1 root root   30126 Mar  5 13:31 [redacted]_ssh_logs.pcap

┌──(kali㉿kali)-[~/Desktop]
└─$ sudo python3 advanced_pcap.py
[sudo] password for kali:
Successfully created a realistic PCAP with 190 packets.

┌──(kali㉿kali)-[~/Desktop]
└─$ la -la
total 6548
drwxr-xr-x  2 kali kali    4096 Mar  5 15:48 .
drwx------ 17 kali kali    4096 Mar  5 13:12 ..
-rw-rw-r--  1 kali kali    2170 Mar  5 15:47 advanced_pcap.py
-rw-rw-r--  1 kali kali     863 Mar  5 13:22 convert2pcap.py
-rw-r--r--  1 kali kali 6595848 Feb 21 13:14 lab1-capture.pcapng
-rw-rw-r--  1 kali kali   21932 Mar  4 22:19 my_ssh_logs.txt
-rw-r--r--  1 root root   30126 Mar  5 15:48 realistic_ssh_logs.pcap
-rw-r--r--  1 root root   30126 Mar  5 13:31 [redacted]_ssh_logs.pcap

┌──(kali㉿kali)-[~/Desktop]
└─$ wireshark
```

Figure 5: PCAP Generation Script Execution

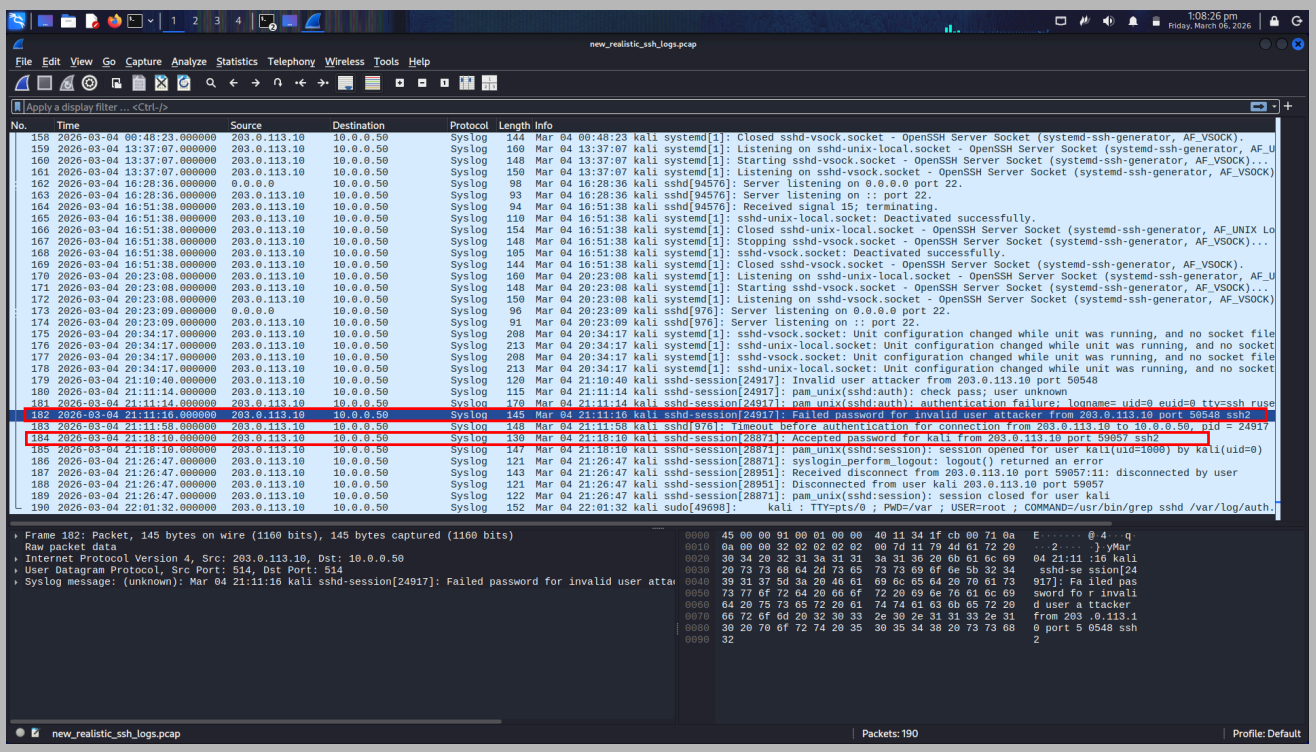

Figure 6: Wireshark Network Traffic Analysis